

$$4.7 \text{ (a)} \quad X_c(j\Omega) = S_c(j\Omega) + \alpha S_c(j\Omega) e^{-j\Omega T_d} \\ = (1 + \alpha e^{-j\Omega T_d}) S_c(j\Omega)$$

$$X(e^{j\omega}) = \frac{1}{T} \sum_{n=-\infty}^{+\infty} X_c(j(\frac{\omega}{T} - \frac{2\pi n}{T})) \\ = \frac{1}{T} \sum_{n=-\infty}^{+\infty} (1 + \alpha e^{-j(\frac{\omega}{T} - \frac{2\pi n}{T})T_d}) S_c(j(\frac{\omega}{T} - \frac{2\pi n}{T}))$$

$$(b) \quad S[n] = S_c(nT) \rightarrow \boxed{D/C} \rightarrow \boxed{K(j\omega)} \rightarrow \boxed{C/D} \rightarrow r[n] = x_c(nT)$$

$$S(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} S_c(j(\frac{\omega}{T} - \frac{2\pi k}{T}))$$

$$H(e^{j\omega}) = 1 + \alpha e^{-j(\frac{\omega - 2\pi \lfloor \frac{\omega + \pi}{2\pi} \rfloor}{T})T_d}$$

$$= 1 + \frac{\alpha}{2\pi} \sum_{k=-\infty}^{+\infty} \int_{-\pi}^{\pi} e^{-j\frac{\omega T_d}{T}} \cdot e^{-j k \omega} d\omega$$

$$= 1 + \alpha \cdot \sum_{k=-\infty}^{+\infty} \frac{\sin[(\frac{T_d}{T} + k)\pi]}{(\frac{T_d}{T} + k)\pi} e^{j k \omega}$$

$$= 1 + \alpha \sum_{k=-\infty}^{+\infty} S_a(\frac{T_d}{T} + k) e^{j k \omega}$$

$$(c) \quad h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} (1 + \alpha e^{-j\frac{\omega T_d}{T}}) e^{j\omega n} d\omega \\ = \frac{1}{2\pi} \cdot \alpha \int_{-\pi}^{\pi} e^{j\omega (n - \frac{T_d}{T})} d\omega + \delta[n]$$

$$(i) \quad T_d = T$$

$$h[n] = \delta[n] + \alpha \delta[n-1]$$

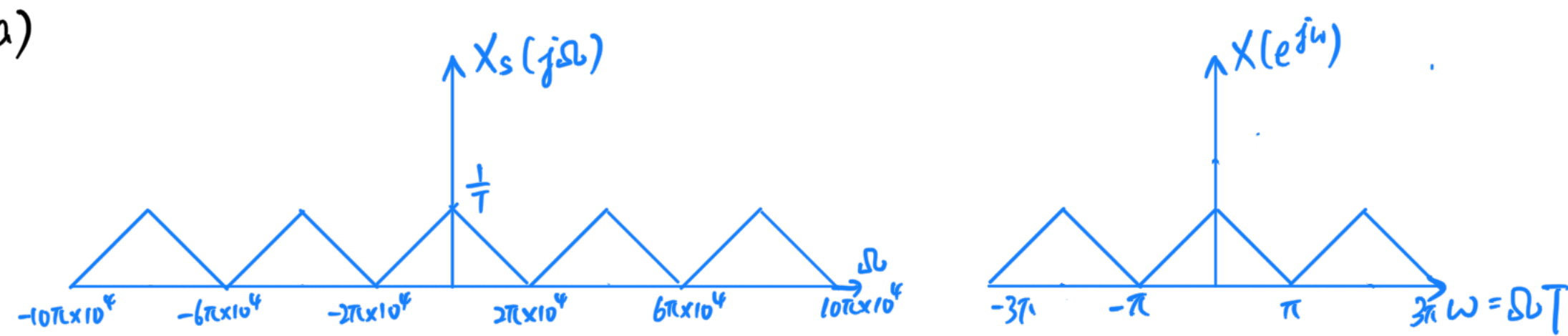
$$(ii) \quad T_d = \frac{T}{2}$$

$$h[n] = \frac{\alpha}{2\pi} \cdot \frac{1}{j(n - \frac{1}{2})} \left(e^{j\pi(n - \frac{1}{2})} - e^{j(-\pi)(n - \frac{1}{2})} \right) + \delta[n]$$

$$= \frac{\alpha}{2\pi} \cdot \frac{2j \sin(\pi(n - \frac{1}{2}))}{j(n - \frac{1}{2})} + \delta[n]$$

$$= \alpha \cdot \frac{\sin(\pi(n - \frac{1}{2}))}{\pi(n - \frac{1}{2})} = \alpha S_a(n - \frac{1}{2}) + \delta[n]$$

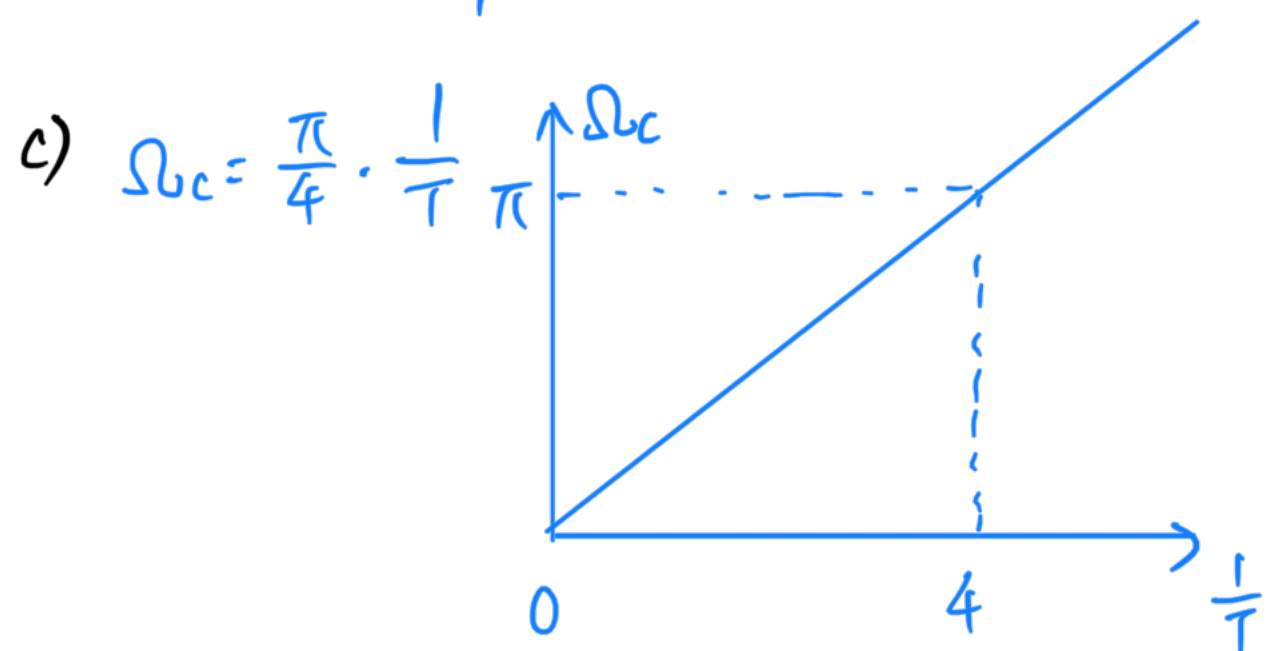
4.31 (a)



$$(b) \quad Y_c(j\Omega) = H_r(j\Omega) Y_c(j\Omega) = H_r(j\Omega) Y(e^{j\Omega T}) = H_r(j\Omega) H(e^{j\Omega T}) X(e^{j\Omega T}) \\ = H_r(j\Omega) H(e^{j\Omega T}) X_s(j\Omega) = H_r(j\Omega) H(e^{j\Omega T}) \cdot \frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\Omega - \frac{2\pi k}{T})) \\ = \begin{cases} 0, & |\Omega| > \frac{\pi}{4T} \\ \sum_{k=-\infty}^{+\infty} X_c(j(\Omega - \frac{2\pi k}{T})), & |\Omega| \leq \frac{\pi}{4T} \end{cases}$$

$$T \text{ 满足 } \frac{2\pi}{T} \geq 2 \times 2\pi \times 10^4 \Leftrightarrow T \leq \frac{1}{2 \times 10^4} \text{ s} = 5 \times 10^{-5} \text{ s} = T_c$$

在 a) 中, $\frac{1}{T} = 20 \text{ kHz} \Leftrightarrow T = 5 \times 10^{-5} \text{ s} = T_c \Rightarrow$ a) 中结论成立



$$4.15 \quad X_r(e^{j\omega}) = H(e^{j\omega}) X_e(e^{j\omega}) = H(e^{j\omega}) X_d(e^{j\omega L}) = \frac{1}{M} H(e^{j\omega}) \sum_{k=0}^{M-1} X(e^{j(\omega - \frac{2\pi k}{M})})$$

$$= \begin{cases} \sum_{k=0}^{M-1} X(e^{j(\omega - \frac{2\pi k}{M})}) & , |\omega| \leq \frac{\pi}{3} \\ 0, & , |\omega| \in (\frac{\pi}{3}, \pi] \end{cases}$$

$$(a) \quad x[n] = \cos(\pi n/4) = \frac{1}{2} (e^{j\frac{\pi}{4}n} + e^{j(-\frac{\pi}{4})n})$$

$$X(e^{j\omega}) = \frac{1}{2} \cdot 2\pi \left(\sum_{k=-\infty}^{+\infty} \delta(\omega - \frac{\pi}{4} - 2\pi k) + \sum_{k=-\infty}^{+\infty} \delta(\omega + \frac{\pi}{4} - 2\pi k) \right)$$

$$\omega_N = \frac{\pi}{4} \leq \frac{\pi}{3} \Rightarrow x_r[n] = x[n]$$

$$(b) \quad x[n] = \cos(\pi n/2)$$

$$\Rightarrow \omega_N = \frac{\pi}{2} > \frac{\pi}{3} \Rightarrow x_r[n] \neq x[n]$$

$$(c) \quad g[n] = \frac{\sin(\pi n/8)}{\pi n/8} \Leftrightarrow G(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{8} \\ 0, & \text{otherwise} \end{cases}$$

$$x[n] = 64 g[n]^2 \Leftrightarrow X(e^{j\omega}) = 64 \cdot \frac{1}{2\pi} G(e^{j\omega}) * G(e^{j\omega}), (|\omega| \leq \pi)$$

$$= \frac{64}{2\pi} \int_{-\pi}^{\pi} G(e^{j\theta}) G(e^{j(\omega - \theta)}) d\theta$$

$$= \frac{64}{2\pi} \int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} G(e^{j(\omega - \theta)}) d\theta$$

$$= \frac{64}{2\pi} \int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} G(e^{j(\theta - \omega)}) d\theta$$

$$= \frac{64}{2\pi} \begin{cases} 0, & |\omega| > \frac{\pi}{4} \\ \frac{\pi}{4} - |\omega|, & |\omega| \leq \frac{\pi}{4} \end{cases}$$

$$\omega_N = \frac{\pi}{4} < \frac{\pi}{3} \Rightarrow x_r[n] = x[n]$$

4.36 (a)

$$x[n] \xrightarrow{\boxed{5\uparrow}} \boxed{X(e^{j\omega})} \xrightarrow{\boxed{3\downarrow}} \boxed{\frac{1}{3} \sum_{k=0}^2 X(e^{j(\omega - \frac{2\pi k}{3})})} \rightarrow y[n]$$

$$(b) \quad x[n] = \delta[n] \Leftrightarrow X(e^{j\omega}) = 1 \Leftrightarrow X(e^{j\omega}) = 1 \Rightarrow \frac{1}{3} \sum_{k=0}^2 X(e^{j(\omega - \frac{2\pi k}{3})}) = 1$$

$$\Rightarrow Y(e^{j\omega}) = \begin{cases} 3, & |\omega| \leq \frac{\pi}{3} \\ 0, & \text{otherwise} \end{cases}$$

$$\Leftrightarrow y[n] = \frac{1}{2\pi} \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} 3 \cdot e^{j\omega n} d\omega$$

$$= \frac{3}{2\pi} \cdot \frac{1}{jn} \cdot 2j \sin(\pi n/3)$$

$$= \frac{\sin(\pi n/3)}{\pi n/3} = S_a(n/3)$$